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Mine Production Dataset â€” NCL

Monthly Extraction, MGR Dispatch & Super Thermal Feeds

CATEGORY	MINE / BLOCK	YEAR	CLASSIFICATION
Production	Northern Coalfields	2024	CONFIDENTIAL

DOCUMENT VERIFICATION & ACCESS REGISTER

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SECTION 1.0 â€” DOCUMENT INFORMATION & CONTROL

Official Technical Record & Geological Data Logging

The Annual Coal Production Report for Fiscal Year 2024-25 synthesizes production, dispatch, stripping, and equipment performance across Eastern Coalfields operational command areas.

All metrics comply with Ministry of Coal guidelines and the Coal Mines Nationalization Act accounting principles.

Corporate Operational Register

Parameter	Operational Record	Auditing Agency	Status
Reporting Period	01 April 2024 to 31 March 2025	Internal Audit Board	Audited
Total Coal Produced	69.42 Million Tonnes (MT)	Govt. Coal Controller	Verified
Total Dispatch	71.05 Million Tonnes (MT)	Railway Freight Division	Reconciled
Target Achievement	102.08% of Annual Target (68.0 MT)	Board of Directors	Exceeded

SECTION 2.0 â€” EXECUTIVE SUMMARY & CORPORATE HIGHLIGHTS

Official Technical Record & Geological Data Logging

Eastern Coalfields recorded its highest ever annual coal production of 69.42 MT during FY 2024-25, representing a year-on-year growth of 8.47% over FY 2023-24 (64.0 MT).

Overburden removal reached 182.4 Million Cubic Meters (M.Cu.m), driven by heavy deployment of 42m3 electric rope shovels and high-capacity dumper fleets.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Dispatch outpaced gross production through strategic drawdown of pit-head inventory, supplying critical thermal power plants with 58.2 MT of coal.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Gross Annual Production:**	69.42 MT	102.1% of target
â€¢ **Total Coal Dispatch:**	71.05 MT	Rail: 82%, MGR: 14%, Road: 4%
â€¢ **Overburden Removal:**	182.4 M.Cu.m	Stripping ratio 2.63 m3/T

SECTION 3.0 â€” ANNUAL PRODUCTION OVERVIEW & TRAJECTORY

Official Technical Record & Geological Data Logging

Production performance accelerated strongly in Q3 and Q4, overcoming monsoonal inundation challenges during July and August 2024.

Quarterly Coal Production & Dispatch Performance

Quarter	Target (MT)	Actual Prod (MT)	Dispatch (MT)	% Achievement
Q1 (Apr - Jun 2024)	15.50 MT	16.12 MT	16.45 MT	104.0%
Q2 (Jul - Sep 2024)	12.80 MT	13.05 MT	13.40 MT	101.9%
Q3 (Oct - Dec 2024)	18.20 MT	18.90 MT	19.60 MT	103.8%
Q4 (Jan - Mar 2025)	21.50 MT	21.35 MT	21.60 MT	99.3%
ANNUAL TOTAL	68.00 MT	69.42 MT	71.05 MT	102.1%

SECTION 4.0 â€” MONTHLY PRODUCTION BREAKDOWN (FY 2024-25)

Official Technical Record & Geological Data Logging

A granular analysis of monthly extraction reveals peak daily production exceeding 265,000 tonnes during March 2025.

Month-wise Extraction Summary

Month	Opencast (MT)	Underground (MT)	Total Prod (MT)	Daily Avg (kT)
April 2024	5.10	0.24	5.34 MT	178.0 kT/day
May 2024	5.25	0.25	5.50 MT	177.4 kT/day
June 2024	5.05	0.23	5.28 MT	176.0 kT/day
July 2024	4.10	0.20	4.30 MT	138.7 kT/day
August 2024	4.00	0.21	4.21 MT	135.8 kT/day
September 2024	4.32	0.22	4.54 MT	151.3 kT/day

SECTION 5.0 â€” AREA-WISE & MINE-WISE PRODUCTION COMPARISON

Official Technical Record & Geological Data Logging

The Rajmahal and Sonepur Bazari opencast mining complexes contributed over 62% of aggregate annual volume.

Major Mining Area Distribution

Operational Area	Mines Active	FY 24-25 (MT)	FY 23-24 (MT)	Growth %
Rajmahal Opencast	2 Mega Pits	24.50 MT	22.80 MT	+7.45%
Sonepur Bazari OC	1 Mega Pit	18.60 MT	16.90 MT	+10.06%
Mugma Area	6 Mixed Mines	8.40 MT	7.95 MT	+5.66%
Kenda & Sripur Area	8 Mixed Mines	10.12 MT	9.45 MT	+7.09%
Salanpur Area	3 Mixed Mines	7.80 MT	6.90 MT	+13.04%

SECTION 6.0 â€” COAL DISPATCH & EVACUATION LOGISTICS

Official Technical Record & Geological Data Logging

Indian Railways supplied an average of 42.5 rakes/day against an indent of 44.0 rakes/day, enabling smooth evacuation without railhead congestion.

Mode-wise Dispatch Share

Evacuation Mode	Dispatched Volume (MT)	Percentage Share	Major Consumers
Indian Railways (Rake)	58.26 MT	82.0%	NTPC Farakka, Kahalgaon, DVC
Merry-Go-Round (MGR)	9.95 MT	14.0%	Dedicated Super Thermal Units
Road Logistics	2.84 MT	4.0%	State Brick & Non-Power Units

SECTION 7.0 â€” PRODUCTION VS DISPATCH RECONCILIATION

Official Technical Record & Geological Data Logging

Pithead coal stocks were reduced from 4.85 MT as on 1 April 2024 to 3.22 MT by 31 March 2025, ensuring coal freshness and preventing spontaneous combustion risks.

VERIFIED PARAMETER REGISTRATION:

â€¢ Opening Pithead Stock:	4.85 MT	As of 01-Apr-2024
â€¢ Gross Annual Addition:	+69.42 MT	Production volume
â€¢ Gross Evacuation Offtake:	-71.05 MT	Dispatch volume
â€¢ Closing Pithead Stock:	3.22 MT	As of 31-Mar-2025

SECTION 8.0 â€” HEAVY EARTH MOVING MACHINERY (HEMM) FLEET

Official Technical Record & Geological Data Logging

The company operated 48 electric draglines and cable shovels, supported by 265 rigid rear dumpers (100T to 240T capacity).

Major HEMM Equipment Census

Equipment Type	Model / Capacity	Fleet Size	Avg Age (Yrs)	OEM
Electric Rope Shovel	P&H 2800XPB / 33 m3	12 Units	6.2 yrs	Komatsu Mining
Hydraulic Excavator	Liebherr R9800 / 42 m3	8 Units	3.8 yrs	Liebherr
Rigid Dump Truck	CAT 793D / 240 Tonne	64 Units	4.1 yrs	Caterpillar
Rigid Dump Truck	BEML BH100 / 100 Tonne	180 Units	7.4 yrs	BEML India
Walking Dragline	W2000 / 24 m3 Boom 96m	4 Units	14.5 yrs	HMC Heavy

SECTION 9.0 â€” EQUIPMENT AVAILABILITY & UTILIZATION INDEX

Official Technical Record & Geological Data Logging

Heavy equipment availability averaged 84.2% across the fleet against the DGMS benchmark of 80.0%.

VERIFIED PARAMETER REGISTRATION:

â€¢ Shovel Fleet Availability:	86.4%	Target: 85%
â€¢ Shovel Fleet Utilization:	72.8%	Target: 70%
â€¢ Dumper Fleet Availability:	82.1%	Target: 80%
â€¢ Dumper Fleet Utilization:	68.5%	Target: 65%

SECTION 10.0 â€” OVERBURDEN REMOVAL & VOLUMETRIC STRIPPING

Official Technical Record & Geological Data Logging

Overburden removal reached 182.40 Million Cubic Meters, creating an advanced exposed coal reserve of 14.2 MT for seamless Q1 FY 25-26 extraction.

Overburden Removal by Equipment Category

Stripping Fleet	Volume Removed (M.Cu.m)	% Share	Avg Haul Distance (km)
Walking Draglines	34.65 M.Cu.m	19.0%	Direct Cast (0 km)
Electric Shovel-Dumper	118.20 M.Cu.m	64.8%	3.45 km (External Dump)
Hydraulic Shovel-Dumper	29.55 M.Cu.m	16.2%	2.10 km (In-pit Backfill)

SECTION 11.0 â€” STRIPPING RATIO ANALYSIS & PIT ECONOMICS

Official Technical Record & Geological Data Logging

The aggregate composite stripping ratio stood at 2.63 m3 of overburden per tonne of coal mined, well within the economic feasibility ceiling of 3.50 m3/T.

VERIFIED PARAMETER REGISTRATION:

â€¢ Rajmahal Opencast:	1.92 m3/T	Highly favorable thick seam
â€¢ Sonepur Bazari Opencast:	2.84 m3/T	Multiple thin seams
â€¢ Mugma Opencast Pit #4:	3.45 m3/T	Deep bench expansion

SECTION 12.0 â€” WORKFORCE PRODUCTIVITY & OUTPUT PER MANSHIFT

Official Technical Record & Geological Data Logging

Overall Output Per Manshift (OMS) reached a company milestone of 14.82 tonnes/manshift in opencast mines and 1.94 tonnes/manshift in underground operations.

Output Per Manshift (OMS) Trend

Fiscal Year	Opencast OMS (T)	Underground OMS (T)	Overall Company OMS (T)
FY 2021-22	11.85 T	1.62 T	7.45 T
FY 2022-23	12.90 T	1.74 T	8.22 T
FY 2023-24	13.75 T	1.82 T	9.10 T
FY 2024-25	14.82 T	1.94 T	10.05 T

SECTION 13.0 â€” OPERATIONAL BREAKDOWN & DOWNTIME ANALYSIS

Official Technical Record & Geological Data Logging

Unplanned downtime was minimized to 7.4% through predictive vibration analytics and oil sampling on high-tonnage shovel drivetrains.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Preventive Maintenance Hours:**	14,280 hrs	Scheduled compliance 96%
â€¢ **Unscheduled Mechanical Breakdown:**	3,840 hrs	Decreased by 18% YoY
â€¢ **Monsoonal Shift Standstills:**	1,420 hrs	High capacity pumps deployed

SECTION 14.0 â€” SPECIFIC ENERGY & FUEL CONSUMPTION METRICS

Official Technical Record & Geological Data Logging

High-speed diesel (HSD) consumption averaged 0.94 liters per cubic meter of composite excavation, achieving the corporate energy conservation target.

Energy Consumption Benchmarks

Energy Resource	Unit Consumption	Benchmark Standard	Cost per Tonne
HSD Diesel	0.94 L / Cu.m	< 1.05 L / Cu.m	INR 86.40 / T
Grid Electricity	18.4 kWh / T	< 22.0 kWh / T	INR 128.80 / T
Explosives (Bulk ANFO)	0.62 kg / Cu.m	< 0.70 kg / Cu.m	INR 44.50 / T

SECTION 15.0 â€” ENVIRONMENTAL COMPLIANCE & LAND RECLAMATION

Official Technical Record & Geological Data Logging

Concurrent biological and technical backfilling reclaimed 245 hectares of mined-out pit area, planted with 620,000 indigenous tree saplings.

VERIFIED PARAMETER REGISTRATION:

â€¢ Technical Reclamation:	**310 Hectares**	Backfilled to original contour
â€¢ Biological Reclamation:	**245 Hectares**	Dense canopy forestry
â€¢ Effluent Treatment Compliance:	**100% Zero Discharge**	Water recycled for dust suppression

SECTION 16.0 â€” FIVE-YEAR PRODUCTION TREND (2020 TO 2025)

Official Technical Record & Geological Data Logging

The multi-year trajectory highlights robust expansion: 2020 (48 MT), 2021 (53 MT), 2022 (59 MT), 2023 (64 MT), 2024 (69 MT), and projected 2025 (74 MT).

5-Year Production vs Dispatch Trajectory (Recharts Grounding)

Year	Production (MT)	Dispatch (MT)	OB Removal (M.Cu.m)	Gross Revenue (Cr)
2020	48.0 MT	45.0 MT	124.0 M.Cu.m	INR 8,420 Cr
2021	53.0 MT	50.0 MT	138.5 M.Cu.m	INR 9,650 Cr
2022	59.0 MT	56.0 MT	152.0 M.Cu.m	INR 11,200 Cr
2023	64.0 MT	61.0 MT	168.0 M.Cu.m	INR 12,850 Cr
2024	69.0 MT	66.0 MT	182.4 M.Cu.m	INR 14,400 Cr
2025 (Est)	74.0 MT	71.0 MT	198.0 M.Cu.m	INR 15,900 Cr

SECTION 17.0 â€” STRATEGIC OPERATIONAL OBSERVATIONS

Official Technical Record & Geological Data Logging

1. Highwall face stability requires real-time radar monitoring during monsoon months at Rajmahal Section II.

2. Rail siding electrification at Sonepur Bazari has reduced rake turnaround times from 4.2 hours to 2.8 hours.

3. Introduction of digital dispatch ticketing eliminated weight reconciliation discrepancies with thermal power utilities.